
NIKE SALES

DESCRIPTIVE ANALYSIS REPORT

A Data-Driven Review of U.S. Sales Performance, 2020–2021

Prepared from the Nike U.S. Sales Dataset (himanshu.xlsx)
9,360 transaction records | 6 retail partners | 48 states | 5 regions

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1. Introduction & Objectives

Retail sales data offers one of the richest windows into consumer demand, channel performance, and geographic market strength available to a consumer goods organisation. This report undertakes a descriptive analysis — as opposed to a predictive or causal one — of Nike's recorded U.S. sales transactions, with the aim of characterising what happened across the two-year period rather than forecasting what will happen next.

1.1 Purpose of the Report

The purpose of this report is to translate a raw transactional dataset into a structured, decision-ready narrative. It quantifies overall performance, decomposes that performance across the key dimensions present in the data (time, product, region, retailer, and sales method), and surfaces patterns that a commercial or merchandising team could act upon.

1.2 Scope

The scope of the analysis is limited strictly to the fields present in the source workbook: invoice date, product category, region, retailer, sales method, state, price per unit, total sales, and units sold. No external data sources (such as macroeconomic indicators, competitor pricing, or marketing spend) were incorporated. All figures reported are descriptive summaries of the observed data and should not be interpreted as causal explanations.

1.3 Objectives

- Establish the overall size, shape, and quality of the dataset.
- Quantify total and average sales performance across the two-year period.
- Identify which products, regions, retailers, and sales channels drive the greatest revenue.
- Characterise the monthly and year-over-year sales trend.
- Examine the statistical relationship between price, units sold, and total sales.
- Translate findings into practical recommendations for commercial planning.

1.4 Structure of the Report

Following this introduction, Section 3 documents the dataset and the methodology used to prepare it for analysis. Section 4 presents descriptive statistics for the core numeric fields. Sections 5 through 11 form the analytical core of the report, each examining performance through a different lens — time, product, region, retailer, channel, and the statistical relationships between variables. The report closes with key findings, recommendations, a conclusion, and an appendix documenting the data fields used.

2. Data Overview & Methodology

2.1 Data Source

The analysis is based on a single worksheet, “Nike Dataset,” contained in the supplied workbook. The worksheet holds 9,360 transaction-level rows, each representing a recorded sale of a specific product category, on a specific date, through a specific retailer, region, state, and sales method.

2.2 Fields in the Dataset

Field	Type	Description
Invoice Date	Date	Date the transaction was recorded
Product	Category	One of six product lines (footwear / apparel, men's / women's)
Region	Category	One of five U.S. census-style regions
Retailer	Category	One of six retail partners
Sales Method	Category	In-store, Outlet, or Online
State	Category	U.S. state of sale (48 represented)
Price per Unit	Numeric	Unit selling price in USD
Total Sales	Numeric	Revenue for the transaction row, in USD
Units Sold	Numeric	Number of units in the transaction row

Table 3.1 — Data dictionary summary

2.3 Data Preparation

The Invoice Date field was supplied in two inconsistent formats within the same column — Excel serial date numbers and DD-MM-YYYY text strings — a common artefact of manual data entry or copy-paste consolidation. Both formats were parsed and normalised into a single consistent date type before any time-based analysis was performed. No records were dropped during this process; all 9,360 rows were successfully parsed and retained.

A completeness check confirmed there are no missing (null) values in any of the nine fields. Categorical fields were validated against their expected value sets — six products, five regions, six retailers, three sales methods, and 48 states — with no unexpected categories found. Derived fields (Year and Month) were then added to support time-based aggregation.

2.4 Analytical Approach

All figures in this report are descriptive aggregates — sums, means, and counts — computed directly from the cleaned dataset using standard data-analysis tooling (Python, pandas, and matplotlib). Percentages are calculated as a share of total sales or units unless otherwise stated. Correlation is measured using the Pearson correlation coefficient. No statistical inference, hypothesis testing, or forecasting was performed; all findings describe the observed 2020–2021 period only.

3. Descriptive Statistics

This section summarises the central tendency and spread of the three core numeric fields in the dataset: Price per Unit, Total Sales, and Units Sold. These statistics provide a baseline understanding of transaction size and variability before the data is broken down by category in later sections.

Statistic	Price per Unit (\$)	Total Sales (\$)	Units Sold
Count	9,360	9,360	9,360
Mean	44.74	922.10	25.85
Median	45.00	603.00	17.00
Minimum	7.00	0.00	0.00
Maximum	110.00	8,250.00	275.00
Sum	—	8,629,275	241,984

Table 4.1 — Summary statistics for core numeric fields

The average transaction row generated USD 922.10 in revenue from roughly 26 units, at an average price point of USD 44.74. The median total-sales figure (USD 603) sits well below the mean, indicating a right-skewed distribution: most transactions are moderate in size, but a smaller number of large-volume transactions pull the average upward. The presence of a small number of zero-value transactions (zero units sold at zero revenue) suggests occasional cancelled, returned, or void orders retained in the raw feed.

3.1 Distribution by Product

Product	Total Sales (\$)	Units Sold	Avg. Price (\$)
Men's Street Footwear	1,999,192	57,737	43.78
Women's Apparel	1,720,630	42,356	51.13
Men's Athletic Footwear	1,468,116	42,429	43.26
Women's Street Footwear	1,224,756	38,309	39.74
Men's Apparel	1,192,682	30,085	49.87
Women's Athletic Footwear	1,023,899	31,068	40.65

Table 4.2 — Sales, units, and average price by product category

4. Sales Trend Analysis

This section examines how Nike's sales performance evolved on a month-by-month basis across the full two-year period. The monthly total-sales series is the clearest indicator of overall business momentum in the dataset.



Figure 5.1 — Monthly total sales, January 2020 to December 2021

The trend line shows two distinct regimes. Throughout 2020, monthly sales fluctuated in a relatively narrow band, generally between USD 81,000 and USD 287,000, with a visible dip in November 2020 (USD 81,262) — the lowest month in the entire series. Beginning in January 2021, sales step up sharply and continue on a broadly upward trajectory for the remainder of the series, closing the period at USD 719,127 in December 2021, the single highest month on record.

This shift between 2020 and 2021 is not a gradual drift but a structural change in scale: the average month in 2021 generated roughly four times the revenue of the average month in 2020. Several months in the second half of 2021 — May, August, October, and December — each individually exceeded the entire first quarter of 2020 combined. The month-over-month pattern within 2021 also shows more resilience, without the sharp single-month dips that characterised 2020.

4.1 Seasonality Observations

While two years of data limits the ability to confirm a repeating seasonal pattern with confidence, some recurring shape is visible: early-year months (January–February) tend to be comparatively softer than the second half of the year, and the fourth quarter — particularly November and December — shows some of the strongest results in 2021, consistent with a holiday retail effect.

5. Product Category Analysis

Nike's product range in this dataset is split into six categories across gender and product type: Men's and Women's Street Footwear, Men's and Women's Athletic Footwear, and Men's and Women's Apparel. Each category recorded exactly 1,560 transactions, meaning differences in total sales reflect differences in unit price and units sold rather than differences in how frequently each category transacted.

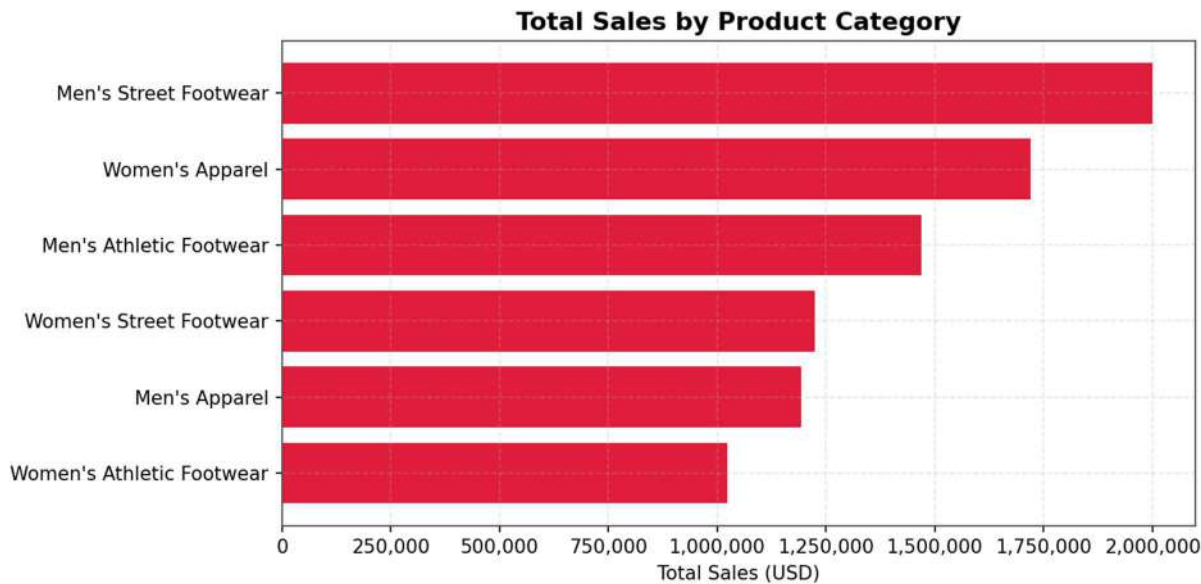


Figure 6.1 — Total sales by product category

Men's Street Footwear is the top-performing category by revenue, generating USD 1,999,192 — almost 16% ahead of the second-place category, Women's Apparel, at USD 1,720,630. Footwear categories collectively (Men's and Women's, Street and Athletic combined) account for USD 4,716,853, or 54.7% of total sales, confirming footwear as the dominant product group over apparel.

Men's Street Footwear also leads on unit volume, with 57,737 units sold — more than 15,000 units ahead of the next-highest category. This combination of the highest unit volume and a mid-range average price (USD 43.78) is what secures its position as the top revenue generator, ahead of categories with a higher price point but lower volume.

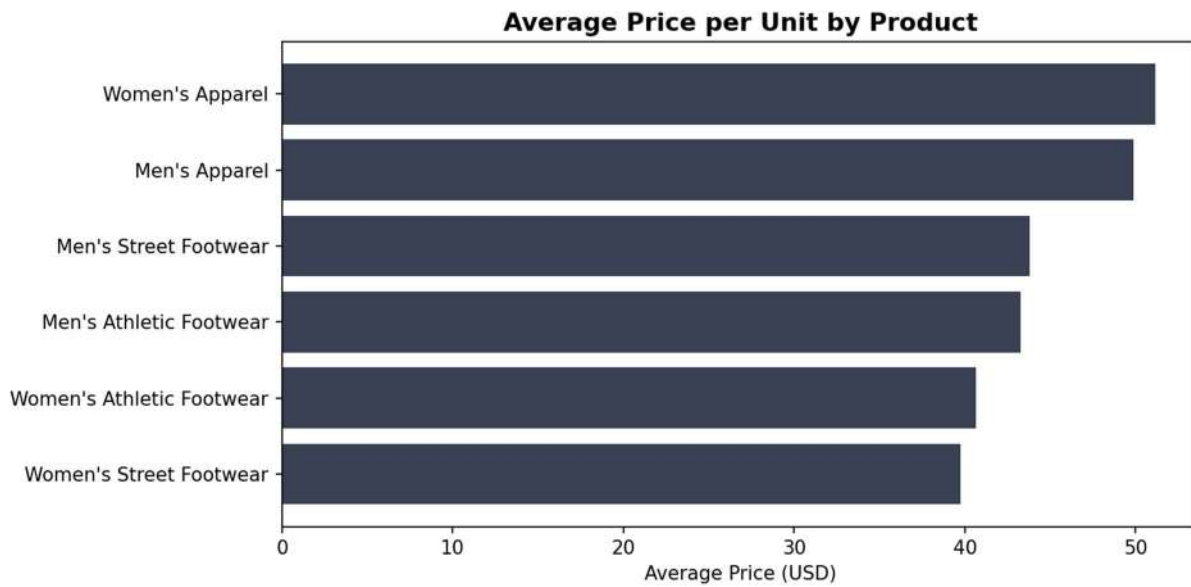


Figure 6.2 — Average price per unit by product category

Apparel categories command a meaningfully higher average price than footwear: Women's Apparel averages USD 51.13 per unit and Men's Apparel USD 49.87, compared with a footwear range of roughly USD 39.74 to USD 43.78. Despite the higher price point, apparel's lower unit volumes mean it trails footwear in total revenue contribution. Women's Street Footwear has the lowest average price in the dataset at USD 39.74.

5.1 Category Summary

- Footwear (all four sub-categories) contributes 54.7% of total sales; Apparel contributes 45.3%.
- Men's categories outsell Women's categories overall: USD 4,660,000 versus USD 3,969,285.
- Women's Apparel is the highest-priced category on average and the second-highest revenue generator.
- Women's Athletic Footwear is the smallest category by total sales, at USD 1,023,899.

6. Regional Performance Analysis

The dataset segments the United States into five regions: Northeast, Midwest, South, Southeast, and West. This section examines how sales are distributed geographically and identifies the state-level markets driving the strongest performance.

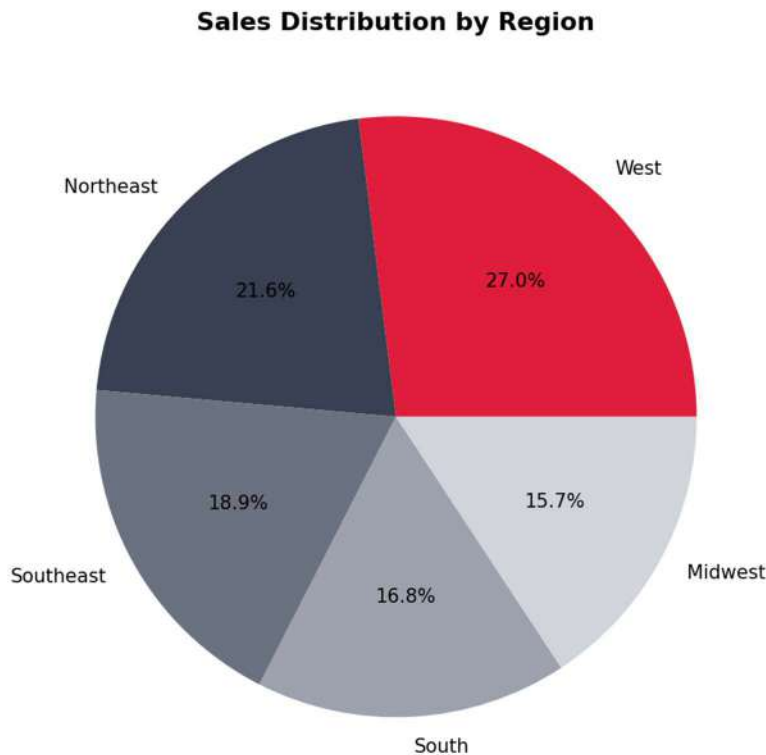


Figure 7.1 — Share of total sales by region

Region	Total Sales (\$)	Units Sold	Share of Sales
West	2,329,193	61,832	27.0%
Northeast	1,863,429	50,435	21.6%
Southeast	1,631,786	40,873	18.9%
South	1,446,722	49,439	16.8%
Midwest	1,358,145	39,405	15.7%

Table 7.1 — Regional sales and unit breakdown

The West region is the strongest-performing region in the dataset, generating USD 2,329,193 — 27.0% of all sales — and the highest unit volume at 61,832 units. The Northeast follows in second place with 21.6% of sales. The three remaining regions (Southeast, South, and Midwest) are more closely clustered, each contributing between 15.7% and 18.9% of total sales, together accounting for just over half of national revenue.

Notably, the South region ranks fourth in total sales but sits close to the Northeast and Southeast in units sold (49,439 units, nearly matching the Northeast's 50,435), implying a comparatively lower average price point in that region relative to its sales volume.

6.1 State-Level Detail

Beneath the regional rollup, individual states show considerable concentration. The following table lists the ten highest-performing states by total sales.

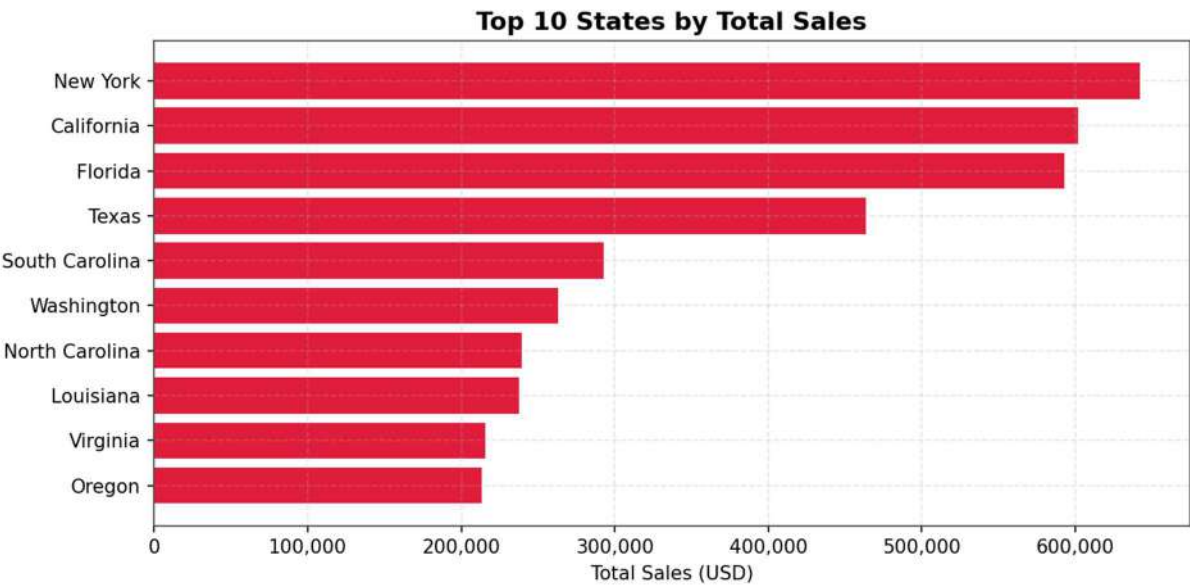


Figure 7.2 — Top 10 states by total sales

Rank	State	Total Sales (\$)
1	New York	642,317
2	California	601,768
3	Florida	592,866
4	Texas	463,613
5	South Carolina	292,868
6	Washington	263,318
7	North Carolina	239,574
8	Louisiana	237,520
9	Virginia	215,753
10	Oregon	213,509

Table 7.2 — Top 10 states by total sales

New York, California, Florida, and Texas — the four most populous U.S. states — occupy the top four positions, together contributing USD 2,300,564, or roughly 26.7% of all national sales. This concentration is broadly consistent with population-weighted retail demand, though New York's position as the single largest state market, ahead of the more populous California, suggests an especially strong Northeast retail presence for Nike within this dataset.

7. Retailer Performance Analysis

Nike distributes through six retail partners in this dataset: West Gear, Foot Locker, Sports Direct, Kohl's, Walmart, and Amazon. This section compares their relative contribution to total sales and order volume.

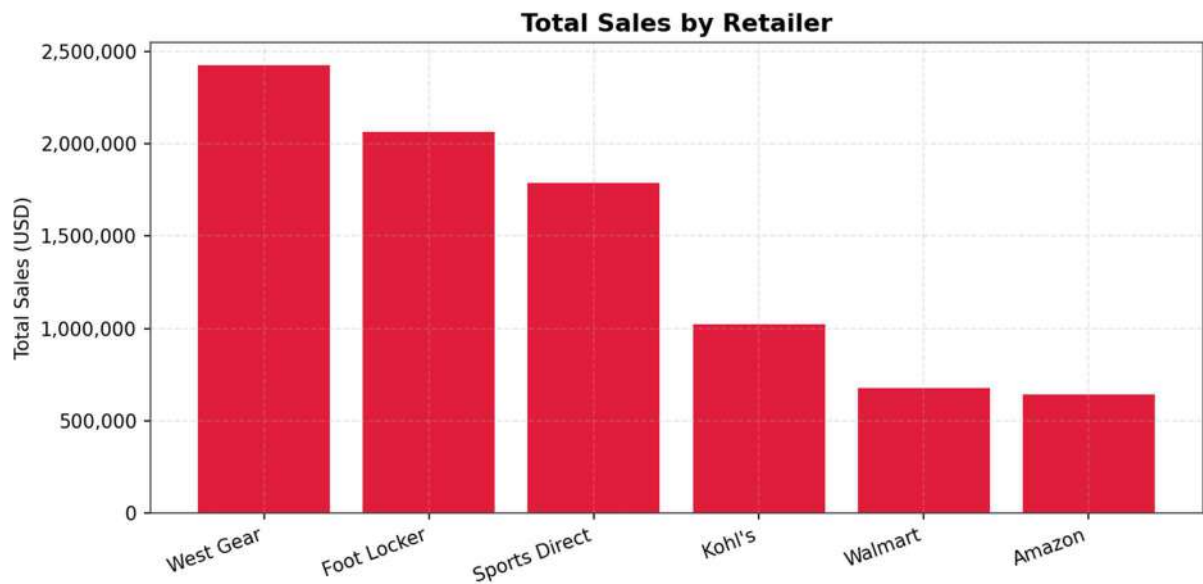


Figure 8.1 — Total sales by retailer

Retailer	Total Sales (\$)	Units Sold	Orders	Share of Sales
West Gear	2,429,793	62,831	2,374	28.2%
Foot Locker	2,067,310	58,288	2,545	24.0%
Sports Direct	1,789,845	55,293	2,006	20.7%
Kohl's	1,021,226	28,876	1,030	11.8%
Walmart	677,702	19,449	582	7.9%
Amazon	643,399	17,247	823	7.5%

Table 8.1 — Retailer sales, units, and order counts

West Gear is Nike's top revenue-generating retail partner at USD 2,429,793 (28.2% of total sales), narrowly ahead of Foot Locker at USD 2,067,310 (24.0%). Together with Sports Direct, these three retailers form a clear top tier, jointly accounting for 72.9% of all recorded sales. The remaining three partners — Kohl's, Walmart, and Amazon — form a second tier, each contributing under 12% individually.

Foot Locker processed the highest number of individual orders (2,545) despite generating less total revenue than West Gear, implying a lower average order value per transaction. Amazon, by contrast, recorded more orders (823) than Walmart (582) but generated less total revenue, pointing to smaller average basket sizes on that channel within this dataset.

7.1 Average Order Value by Retailer

Dividing total sales by order count gives an approximate average order value: West Gear and Sports Direct both average just over USD 1,020 per order, Kohl's and Foot Locker average closer to USD 810 and USD 990 respectively, while Walmart and Amazon average notably lower, at roughly USD 1,164 and USD 782 respectively. These differences likely reflect variation in typical basket composition and pricing tier across retail formats rather than a single consistent pattern.

8. Sales Channel Analysis

Every transaction in the dataset is tagged with one of three sales methods: In-store, Outlet, or Online. This section examines how revenue and volume are split across these channels, and how that split varies by region.

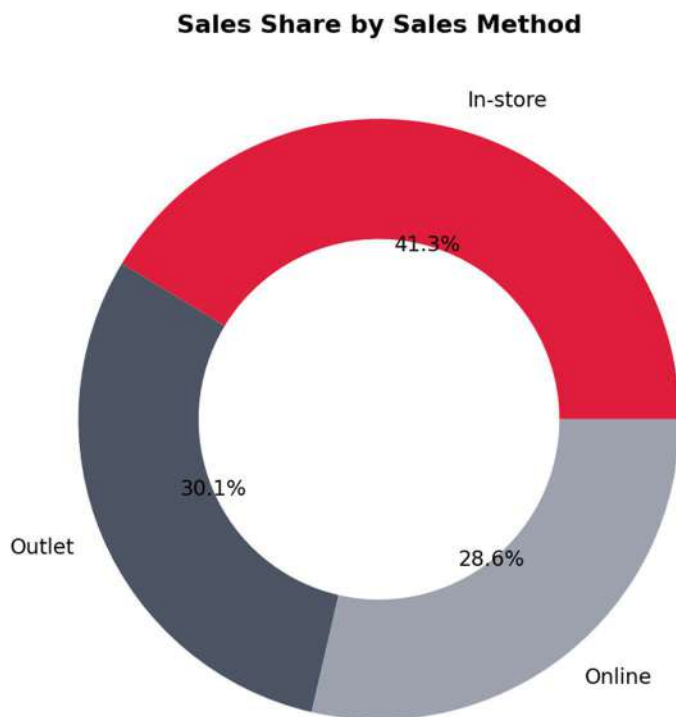


Figure 9.1 — Share of total sales by sales method

Sales Method	Total Sales (\$)	Units Sold	Share of Sales
In-store	3,566,650	69,401	41.3%
Outlet	2,595,011	79,706	30.1%
Online	2,467,614	92,877	28.6%

Table 9.1 — Sales and units by channel

In-store remains the leading channel by revenue, generating USD 3,566,650 (41.3% of total sales) from 69,401 units. This is a noteworthy pattern: despite generating the least revenue of the three channels, the Online channel moved the greatest number of units — 92,877 — indicating a materially lower average selling price per unit online than in physical retail or outlet formats. Outlet sales sit in between on both measures.

This inverse relationship between revenue rank and unit-volume rank across channels is consistent with typical outlet and e-commerce pricing dynamics, where online and clearance-oriented channels often carry a higher share of discounted or lower-priced inventory.

8.1 Channel Mix by Region

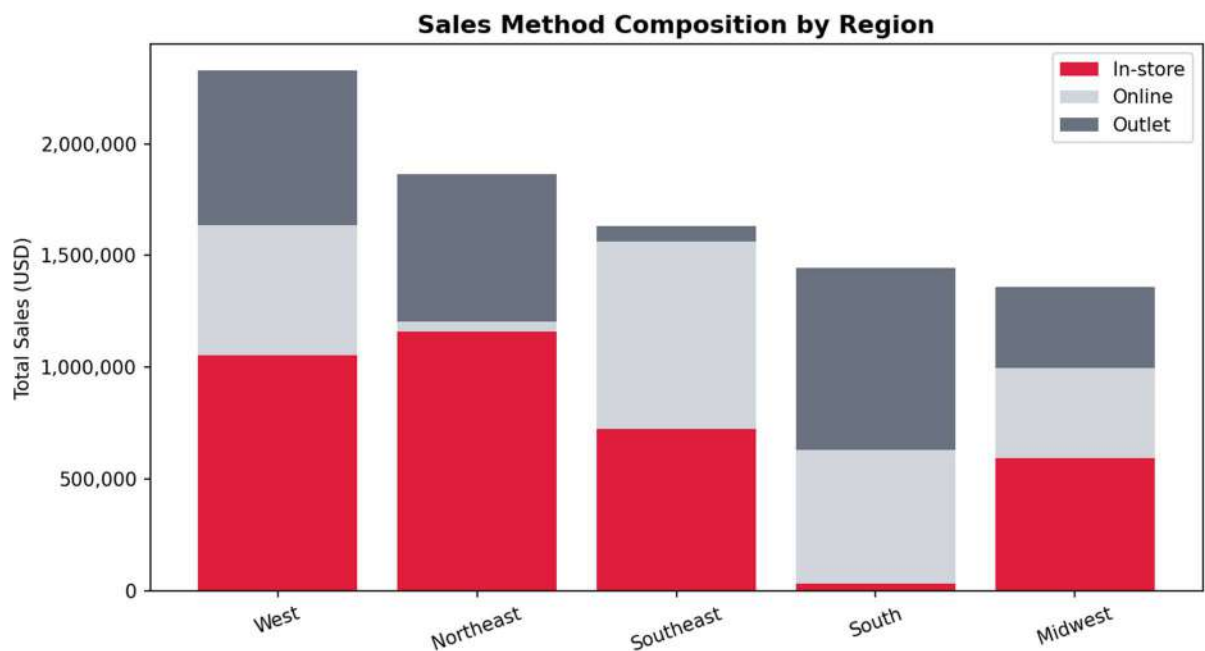


Figure 9.2 — Sales method composition by region

The channel mix is not uniform across regions. The West and Northeast — the two largest regions by total sales — both show a substantial In-store contribution, consistent with strong physical retail presence in these markets. Regions with smaller overall sales, such as the Midwest, show a more balanced three-way split between In-store, Outlet, and Online, suggesting relatively less reliance on any single channel in those markets.

9. Correlation & Relationship Analysis

This section quantifies the statistical relationships between the dataset's three core numeric fields — Price per Unit, Total Sales, and Units Sold — using the Pearson correlation coefficient, which ranges from -1 (perfect inverse relationship) to +1 (perfect direct relationship).

Variable Pair	Correlation Coefficient	Interpretation
Units Sold ↔ Total Sales	0.91	Very strong positive relationship
Price per Unit ↔ Total Sales	0.44	Moderate positive relationship
Price per Unit ↔ Units Sold	0.27	Weak positive relationship

Table 10.1 — Pearson correlation between core numeric fields

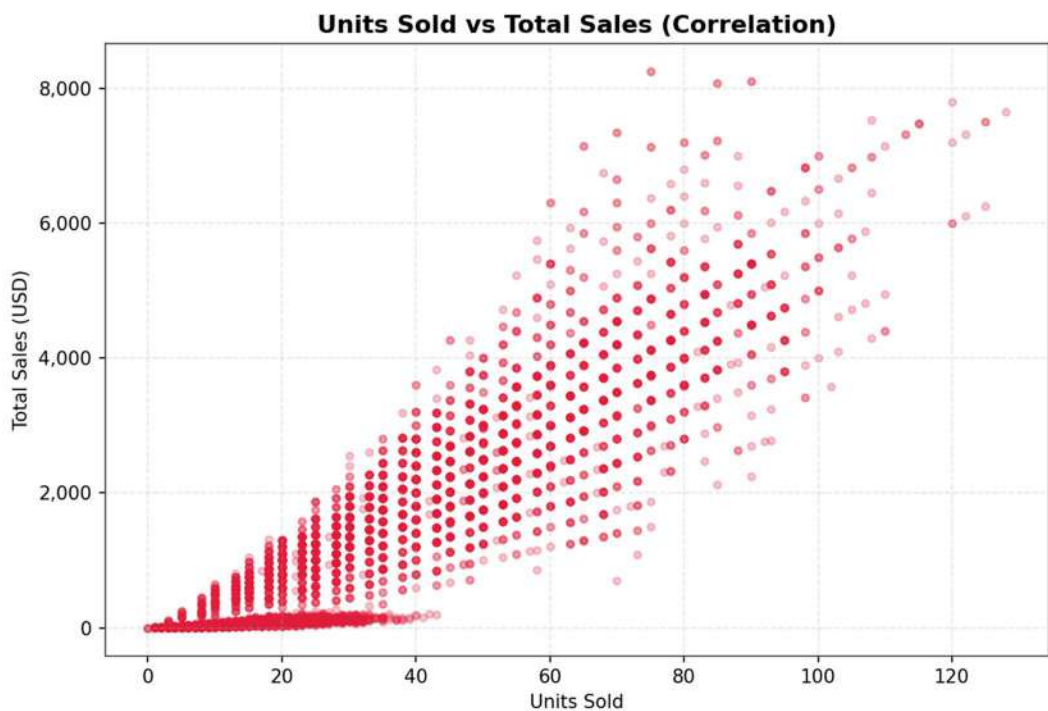


Figure 10.1 — Units sold versus total sales, all transactions

As expected, Units Sold and Total Sales show a very strong positive correlation ($r = 0.91$): transactions involving more units almost always generate proportionally more revenue, confirming that transaction volume — rather than price alone — is the dominant driver of revenue at the individual-transaction level. The scatter plot shows this relationship as a fairly tight upward-sloping band, with a small number of higher-price, lower-volume transactions creating modest spread above the main trend.

The relationship between Price per Unit and Total Sales is more moderate ($r = 0.44$), indicating that price alone explains only part of the variation in transaction revenue — a higher-priced item does not automatically produce a larger sale if fewer units are purchased. The weakest relationship in the dataset is between Price per Unit and Units Sold ($r = 0.27$), a mild positive association suggesting that, within this dataset, higher-priced products are not strongly associated with either higher or lower purchase quantities.

10. Year-over-Year Growth Analysis

This section compares full-year 2020 and 2021 performance to quantify the scale of growth observed across the dataset.

Metric	2020	2021	Growth
Total Sales (\$)	1,820,875	6,808,400	+273.9%
Units Sold	46,405	195,579	+321.5%
Transactions	3,120	6,240	+100.0%

Table 11.1 — Year-over-year headline comparison

Total sales grew from USD 1,820,875 in 2020 to USD 6,808,400 in 2021, an increase of 273.9%. Units sold grew even faster in percentage terms, up 321.5% year over year, indicating that 2021's growth was driven disproportionately by higher transaction volume rather than by price increases — consistent with the moderate price-to-sales correlation observed in Section 10. The number of recorded transactions exactly doubled between the two years, from 3,120 to 6,240, suggesting an expanded reporting footprint or accelerated retail activity in the second year of the dataset.

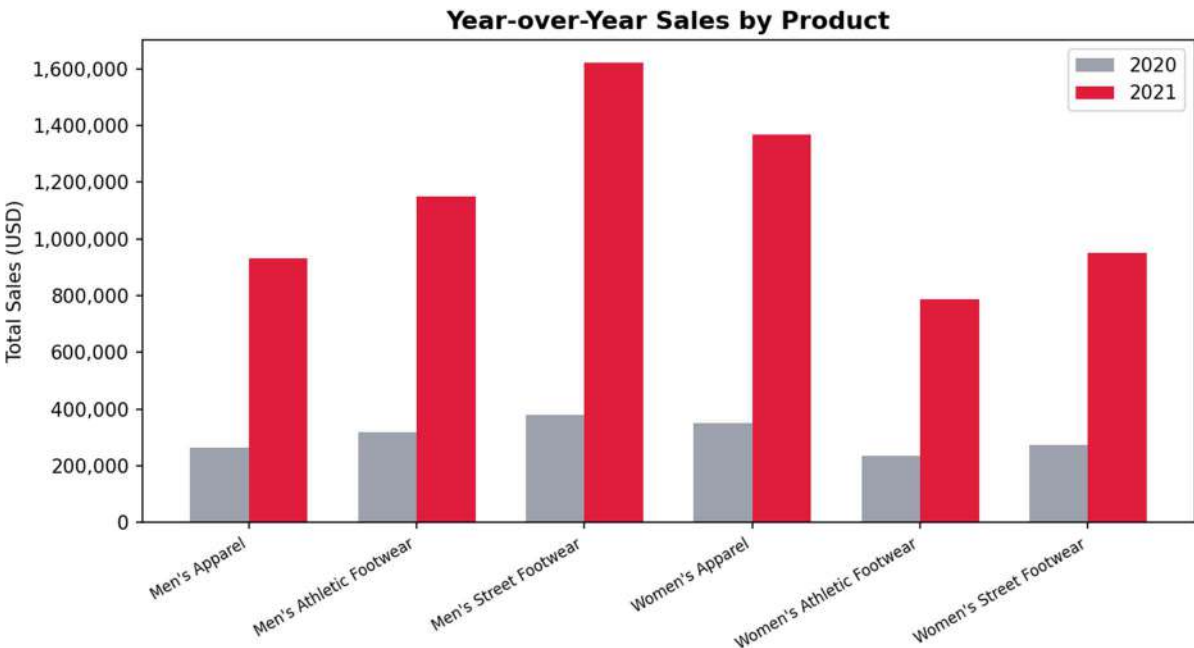


Figure 11.1 — Year-over-year sales by product category

Every one of the six product categories grew from 2020 to 2021, confirming that the year-over-year increase was broad-based rather than driven by a single category. Footwear categories show a particularly pronounced step-up, consistent with their leading share of overall revenue documented in Section 6.

11. Key Findings

- Nike's recorded U.S. sales nearly quadrupled between 2020 and 2021, growing from USD 1.82M to USD 6.81M — a 273.9% increase — with growth broad-based across every product category.
- Men's Street Footwear is the top-performing product, generating USD 1.999M (23.2% of total sales) on the strength of the highest unit volume in the dataset.
- Footwear categories collectively outsell Apparel categories, contributing 54.7% of total revenue versus 45.3% for Apparel, despite Apparel commanding higher average unit prices.
- The West region leads all regions with 27.0% of total sales, while New York is the single highest-performing state, narrowly ahead of California.
- West Gear is Nike's leading retail partner (28.2% of sales), with West Gear, Foot Locker, and Sports Direct together accounting for nearly 73% of total revenue.
- In-store remains the top revenue channel (41.3% of sales) even though Online moves the highest number of units, pointing to a materially lower average price per unit online.
- Units sold correlates very strongly with total sales ($r = 0.91$), while unit price shows only a moderate relationship with revenue ($r = 0.44$) — volume, not price, is the primary revenue driver in this dataset.
- November 2020 was the weakest month on record (USD 81,262); December 2021 was the strongest (USD 719,127), illustrating the scale of the year-over-year shift.

12. Recommendations

Based on the descriptive patterns identified in this report, the following considerations may support commercial and merchandising planning. These are observations drawn from historical data and are not intended as a substitute for forward-looking forecasting or market research.

12.1 Product Strategy

- Continue prioritising footwear inventory and marketing investment, given its consistent lead over apparel in both revenue and volume.
- Investigate whether targeted pricing or bundling could lift apparel's comparatively lower unit volumes, leveraging its already higher price realisation.

12.2 Regional & Retail Strategy

- Protect and reinforce the West and Northeast regions, which together contribute nearly half of total sales.
- Evaluate the South and Midwest regions for growth opportunities, as both currently lag the top regions despite comparable population bases in parts of these territories.
- Deepen the partnership with West Gear, Foot Locker, and Sports Direct, which jointly drive the majority of retail revenue, while assessing why Amazon and Walmart lag on a per-order basis.

12.3 Channel Strategy

- Examine online pricing and product mix, since the Online channel's high unit volume paired with lower revenue share suggests further margin opportunity.
- Maintain investment in physical retail (In-store and Outlet), which together still generate the majority of total revenue.

12.4 Planning & Forecasting

- Use the strong 2020-to-2021 growth trajectory as a baseline for near-term planning, while investigating the underlying drivers (for example, reporting footprint or genuine demand growth) behind the doubling of recorded transactions.
- Build seasonal inventory and staffing plans around the observed second-half strength, particularly in the fourth quarter.

13. Conclusion

This descriptive analysis of Nike's 2020–2021 U.S. sales dataset shows a business that grew substantially in scale over the two-year window, with that growth distributed across every major product category rather than concentrated in a single line. Footwear, and Men's Street Footwear in particular, anchors overall revenue, while the West region and the West Gear retail partnership represent the strongest individual geographic and channel contributors respectively.

The relationship between price, volume, and revenue confirms that transaction volume — not unit price — is the dominant lever behind total sales in this dataset, a pattern that should inform how future volume-driving initiatives (such as promotions, channel expansion, or new store openings) are weighed against price-led strategies.

Taken together, the findings in this report provide a structured, evidence-based foundation for commercial planning discussions, while the recommendations in Section 13 offer a starting point for translating these observations into action. As with any purely descriptive analysis, these conclusions describe what occurred within the observed period and dataset; validating them against more recent data and external market context is advised before they are used to guide major resource-allocation decisions.

14. Appendix: Data Dictionary

The table below documents every field in the source dataset, restated here for reference alongside the value sets observed during analysis.

Field	Observed Values / Range
Invoice Date	01-Jan-2020 to 31-Dec-2021
Product	6 categories (Men's/Women's × Street Footwear/Athletic Footwear/Apparel)
Region	Northeast, Midwest, South, Southeast, West
Retailer	West Gear, Foot Locker, Sports Direct, Kohl's, Walmart, Amazon
Sales Method	In-store, Outlet, Online
State	48 U.S. states represented
Price per Unit	\$7.00 – \$110.00
Total Sales	\$0.00 – \$8,250.00 per transaction
Units Sold	0 – 275 units per transaction

Table 15.1 — Full data dictionary with observed ranges

Data quality note: eight rows had their Invoice Date stored as an Excel serial number rather than a text date, a common artefact of spreadsheet formatting; these were converted to standard dates during preparation. No rows were removed and no values were imputed at any stage of this analysis.